

Setup Guide: Perimeter Patrol

Time Allocation: 45 Minutes

Challenge Overview

The robot must follow the outer border of a designated 2-meter square taped boundary for two complete loops.

Environment & Arena Setup

- **Arena Preparation:** Clear a flat, smooth indoor floor space (such as a gymnasium or tiled classroom floor) free of major debris or severe surface cracks.
- **Boundary Creation:** Using high-visibility floor tape (contrasting with the floor color, such as black electrical tape on a white floor, or vice versa), mark a precise 2-meter by 2-meter square.
- **Precision Check:** Ensure the corners of the square form accurate 90-degree angles and that the tape is pressed down firmly so sensors do not catch on loose edges.
- **Starting Zone:** Designate one specific corner or side of the square as the official starting and finishing line, clearly marking it with a secondary piece of tape to help students align their robots consistently.